Jessica E. Snyder, PhD

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Engineer and explorer seeking an ambitious team to build, as much as evolve, biological industrial design

Work Experience

Blue Marble Space Institute of Science, ***Senior Research Investigator*** (2023-present) *Co-Investigator to engineer an astropharamcy* that produces peptide pharmaceuticals.

Odyssey SpaceWorks, ***Cofounder/Head of Science*** (2022-2023) *Built an automated R&D instrument launched to low Earth orbit aboard SpaceX Transporter 9 in November 2023.* The module supervised the growth of insect muscle cells for applications to space-grown biomaterials.

RoosterBio, ***Bioprocessing Application Scientist*** (2021-2022) *Accelerate human mesenchymal stromal cell product and process development.* Proved technical guidance to academics and industry on proper use of well-defined cellular therapy products, including manufacturing for clincal trials.

Universities Space Research Association, contracted to **NASA Ames Research Center** ***Scientist + Synthetic Biology Task Lead*** (2017-2020) *Biological industrial design to outfit human missions to space.* Built biotechnology alongside evolutionary and synthetic biologists to reverse engineer life from extreme environments. Projects included: Digitizing a sea sponge to 3D-print a living filter; a portable device to make protein-based drugs; and field work in Antarctica.

Massachusetts Institute of Technology, Senseable City Lab, ***Postdoctoral Fellow*** (2015-2017) *Audit public health by measuring the virus, bacteria, and chemicals in wastewater collected neighborhood by neighborhood.* Our Underworlds team designed and built a robotic sampling network deployed to the sewers. Geolocated samples were lab analyzed to heatmaps human health and consumption. Instances of the technology were deployed in Boston and Kuwait.

Drexel University, Mechanical Engineering Department ***Research Fellow / PhD Candidate*** (2009-2014) *Developed manufacturing processes to build a model of human liver for space flight medicine.* Used scaffold-guided tissue engineering to reproduce the anatomical, multi-cell type structure of hepatic tissue in collaboration with NASA Johnson Space Center.

Inventions

Heterogeneous filaments, methods of producing the same, scaffolds, methods of producing the same, droplets, ... WO/2017/019300 (2017) https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2017019300

Compositions and Methods for Functionalized Patterning of Tissue Engineering Substrates Including Bioprinting Cell-Laden Constructs for Multicompartment Tissue Chambers. U.S. Patent 12 872,992 (2011) https://www.freepatentsonline.com/y2011/0136162.html

Service

Montefiore Health System, New York City
Collaborator (2016-present) *Joined a clinical trial of virtual reality as a non-pharmacological pain management strategy.* We collect physiological feedback using wearables to generate immersive experiences that hold promise for guiding pediatric patients towards calmer emotional states.

Education
2009 BS/MS Mech Eng, Drexel University
2014 PhD Mech Eng, Drexel University
2015-17 Postdoc Urban Planning, MIT

Adjunct Faculty
2019 Santa Clara University
2021 University of Silicon Valley
2025 Blue Marble Space Young Scientist Program, Mentor

Training
2010 National Space Biomedical Research Institute, NASA Johnson Space Center
2016 National University of Singapore, Singapore-MIT Alliance for Research and Technology
2016 World Health Summit, New Voices
2019 Quixote Expeditions, Antarctica South Shetland Islands, Guest Scientist
2020 Hawaii Space Exploration Analog and Simulation (HI-SEAS), Analog Astronaut
2025 Autodesk Industry Research Network, Resident

Funded Grants - Current Support

1. *An Off Planet Laundromat.* NASA Mars Campaign Office (MCO) New Starts 2026. Lynn J. Rothschild, Jessica Snyder, Kyle Grant-Talbot, Debbie Senesky, Katya Arquilla. $13million/6 yrs
2. *Mycotecture Off Planet.* NASA Innovative Advanced Concepts (NIAC) Phase II 2021. Lynn J. Rothschild, Christopher Maurer, Debbie Senesky, Ivan Gláucio Paulino Lima, Jim Head, Jessica Snyder, Maikel Rheinstadter, Martyn Dade-Robertson. $250k / 2 years
3. *An Astropharmacy.* NASA Innovative Advanced Concepts (NIAC) Phase II 2023. Lynn J. Rothschild (PI), Kate Adamala, David Loftus, Mike Jewett, Jessica Snyder. $250k / 2 years

Scopus Metrics Overview	**h-index**	**Citations**	**Authored Documents**
https://www.scopus.com/56007982800	12	528	24

Peer Reviewed Publications

1. Blum RC, Omrani D, Kunitskaya A, Snyder JE, Bui C, Williams PM, Loftus DJ, Rothschild LJ. The database of peptide medications for human deep space missions demonstrates potential for on-demand" Astropharmacy". Frontiers in Space Technologies.;6:1692554. https://doi.org/10.3389/frspt.2025.1692554
2. Averesch NJ, Berliner AJ, Nangle SN, Zezulka S, Vengerova GL, Ho D, Casale CA, Lehner BA, Snyder JE, Clark KB, Dartnell LR. Microbial biomanufacturing for space-exploration—what to take and when to make. Nature Communications. 2023 Apr 21;14(1):2311. https://www.nature.com/articles/s41467-023-37910-1
3. Zajkowski T, Lee MD, Mondal SS, Carbajal A, Dec R, Brennock PD, Piast RW, Snyder JE, Bense NB, Dzwolak W, Jarosz DF. The hunt for ancient prions: archaeal prion-like domains form amyloid-based epigenetic elements. Molecular Biology and Evolution. 2021 May;38(5):2088-103. https://doi.org/10.1093/molbev/msab010
4. Lehner BA, Schlechten J, Filosa A, Pou AC, Mazzotta DG, Spina F, Teeney L, Snyder JE, Tjon SY, Meyer AS, Brouns SJ. End-to-end mission design for microbial ISRU activities as preparation for a moon village. Acta Astronautica. 2019 Sep 1;162:216-26. https://doi.org/10.1016/j.actaastro.2019.06.001
5. Snyder JE, Walsh D, Carr PA, Rothschild LJ. A makerspace for life support systems in space. Trends in biotechnology. 2019 Nov 1;37(11):1164-74. https://doi.org/10.1016/j.tibtech.2019.05.003
6. Liu Y, Hamid Q, Snyder JE, Wang C, Sun W. Evaluating fabrication feasibility and biomedical application potential of in situ 3D printing technology. Rapid Prototyping Journal. 2016 Oct 17. https://doi.org/10.1108/RPJ-07-2015-0090
7. Snyder JE, Hamid Q, Sun W. Fabrication of microfluidic manifold by precision extrusion deposition and replica molding for cell-laden device. Journal of Manufacturing Science and Engineering. 2016 Apr 1;138(4). https://doi.org/10.1115/1.4031551
8. Snyder JE, Son AR, Hamid Q, Wu H, Sun W. Hetero-cellular prototyping by synchronized multi-material bioprinting for rotary cell culture system. Biofabrication. 2015 Dec 29;8(1):015002. https://doi.org/10.1088/1758-5090/8/1/015002
9. Snyder JE, Son AR, Hamid Q, Wang C, Lui Y, Sun W. Mesenchymal stem cell printing and process regulated cell properties. Biofabrication. 2015 Dec 22;7(4):044106. https://doi.org/10.1088/1758-5090/7/4/044106
10. Snyder JE, Hunger PM, Wang C, Hamid Q, Wegst UG, Sun W. Combined multi-nozzle deposition and freeze casting process to superimpose two porous networks for hierarchical three-dimensional microenvironment. Biofabrication. 2014 Jan 15;6(1):015007. https://doi.org/10.1115/ISFA2012-7105
11. Snyder JE, Hamid Q, Wang C, Chang R, Emami K, Wu H, Sun W. Bioprinting cell-laden matrigel for radioprotection study of liver by pro-drug conversion in a dual-tissue microfluidic chip. Biofabrication. 2011 Sep 1;3(3):034112. https://doi.org/10.1088/1758-5082/3/3/034112